

DUCK HUNT GAME

Using Arduino And Python

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This is a shooting game based on pygame, using Arduino and an accelerometer sensor. In this project a gun pointer is made on the screen, which is moved along x and y coordinates of the accelerometer, and a switch is used to fire at the target.

The gun pointer is aimed by tilting the hardware (accelerometer). A push-to-on switch (S1) is pressed (pulled like a trigger) to shoot and send a signal to the PC that a shot has been fired. The gun pointer is moved towards the bird, and the bird is killed if the positions coincide.

Arduino and Python using pyserial package. Python version used in this project is 3.4.4. Details on Python packages are given in the software section.

Circuit and working

Circuit diagram of the duck hunt game using Arduino Uno and Python is shown in Fig. 2. The x and y pins of the accelerometer are connected to analogue pins A0 and A1 of Arduino. The accelerometer can have different values according to different manufactures. The x and y analogue values are displayed on the serial monitor of Arduino IDE.

Tilt your hardware and note down the minimum and maximum values for x and y. Put these values in Arduino code. This will be mapped from 0-20 in the program. Then, send serial data to Python and press S1 to connect it to pin 2 of Board1, which is used to fire the gun. The accelerometer is powered by 5V pin of the Arduino board. The accelerometer used for this project is shown in Fig. 3, and the author's prototype is shown in Fig. 4.

Software

Python packages. Download the packages before compiling Python program (duck_hunt.py). These can be downloaded by typing 'unofficial windows binaries Python extension packages' on Google. Download 3.4.4 version of Python.

Python packages used in the program are pyserial, pygame, random, time and math, of which you would only have to download pyserial and pygame; the rest are pre-installed.

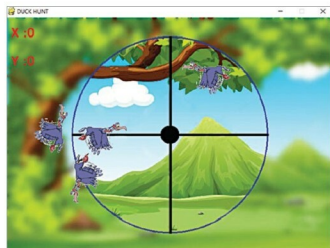


Fig. 1: Screenshot of the duck hunt game

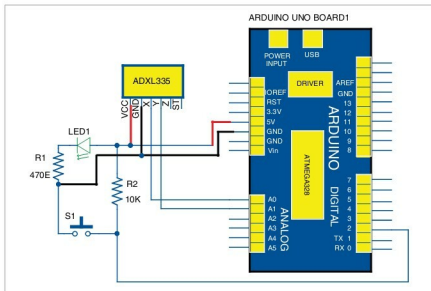


Fig. 2: Circuit diagram of the duck hunt game using Arduino Uno